PROJECT 05: Multi-Plan Telecommunications Cost Optimization

1. Problem Statement

Compare mobile plans against expected data and call usage and a spending limit, including excess-use charges, to find the cheapest eligible plan.

2. Objectives

• Represent plan prices and allowances as data.
• Calculate excess data and call charges.
• Check each plan against a user budget.
• Recommend the lowest-cost plan that meets the budget.

3. Assumptions

• All plans cover one month.
• Data usage is measured in GB and call usage in minutes.
• Excess data charges are per GB; excess call charges are per minute.
• Plan prices and rates are illustrative and do not represent current carrier offers.

4. Algorithm

1. Read usage and budget.
2. For each plan, calculate excess data and call usage.
3. Add excess charges to the monthly plan price.
4. Display each estimated plan cost.
5. Keep plans whose total cost does not exceed the budget.
6. Select the lowest-cost eligible plan or report none.

5. Flowchart

6. Pseudocode

```
START
  READ data_usage, call_minutes, budget
  SET eligible_plans = empty list
  FOR each plan
    extra_data = max(0, data_usage - data_allowance)
    extra_calls = max(0, call_minutes - call_allowance)
    cost = monthly_fee + extra_data * data_rate
           + extra_calls * call_rate
    DISPLAY plan and cost
    IF cost <= budget: ADD plan to eligible_plans
  IF eligible_plans is empty: DISPLAY no eligible plan
  ELSE DISPLAY eligible plan with minimum cost
END
```

7. Python Code

```
plans = [
    {"name": "Basic", "monthly": 199, "data_gb": 2, "call_mins": 300, "extra_gb": 50,
"extra_min": 0.50},
    {"name": "Smart", "monthly": 299, "data_gb": 5, "call_mins": 700, "extra_gb": 40,
```

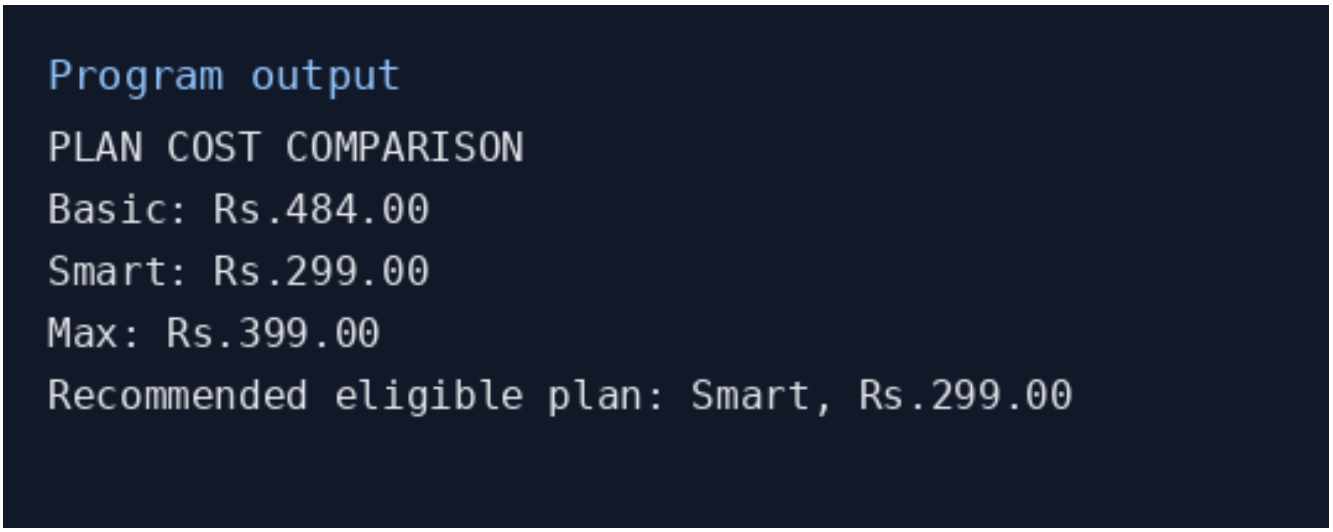
```
"extra_min": 0.40},
    {"name": "Max", "monthly": 399, "data_gb": 10, "call_mins": 1200, "extra_gb": 30,
"extra_min": 0.30},
]
usage_gb, usage_mins, budget = 4.2, 650, 350
options = []
print("PLAN COST COMPARISON")
for p in plans:
    extra_data = max(0, usage_gb - p["data_gb"])
    extra_mins = max(0, usage_mins - p["call_mins"])
    cost = p["monthly"] + extra_data * p["extra_gb"] + extra_mins * p["extra_min"]
    print(f"{p['name']}: Rs.{cost:.2f}")
    if cost <= budget:
        options.append((cost, p["name"]))
if options:
    cost, name = min(options)
    print(f"Recommended eligible plan: {name}, Rs.{cost:.2f}")
else:
    print("No plan meets the budget.")
```

8. Test Cases

Test	Input / condition	Expected result
Within allowance	Usage 1 GB, 200 minutes	No excess charges
Excess data	Usage 4.2 GB on Basic plan	2.2 GB excess is charged
Budget met	Budget Rs.350 with sample usage	Smart is the cheapest eligible plan
No eligible plan	Budget Rs.100	Program reports no plan meets budget

9. Output Screenshot

The following screenshot-style image was generated from the captured output of the program execution.



Captured output

```
PLAN COST COMPARISON
Basic: Rs.484.00
Smart: Rs.299.00
Max: Rs.399.00
Recommended eligible plan: Smart, Rs.299.00
```

10. Conclusion

The program compares plans on estimated total cost, not just monthly sticker price, and respects the user's budget limit.

Academic note: The program was executed with the sample data shown in the source code. Policies, rates, thresholds and exchange rates are illustrative and should be adjusted to the assignment's official requirements.